

EXPERIMENT NO. 20 STUDY OF GUNN OSCILLATOR

Objective:

- (a) To study I-V characteristics of GUNN diode
- (b) To use GUNN diode as a source of amplitude modulated electromagnetic waves in the microwave region and study its characteristics.
- (c) To study the characteristics of PIN diode modulator.
- (d) To study the response law of the detector diode.
- (e) To calibrate the micrometer attenuator in dB.

Theory of GUNN Diode

In 1963, J.B. Gunn discovered that microwave oscillations occur in a simple slab of the GaAs semiconductor under an electric field of several kVcm^{-1} [1]. The oscillations are a consequence of the negative differential mobility (NDM) of the conduction electrons in GaAs. This effect is known as Gunn effect. Similar effect is observed in several other III-V semiconductors such as InP and CdTe.

The occurrence of bulk NDM in III-V semiconductors results due to a typical band structure of these compounds. In these compounds there exist two conduction bands:

- (a) lower energy, high mobility and
- (b) higher energy, lower mobility.

The lower conduction band is called the light mass (effective) band and the upper band is called the heavy mass band. The current density J is the sum of the contributions from the two conduction bands:

$$J = \sigma_{(E)} \vec{E} = (n_l \mu_l + n_h \mu_h) e E \quad (1)$$

where suffix l refers to light mass band and h refers to the heavy mass band, n refers to number of conduction electrons, μ refers to mobility and E is the electric field, and $(n_l + n_h) = N$ (Constant).

The differential conductivity is then given by

$$\frac{dJ}{dE} = (n_l \mu_l + n_h \mu_h) e + \left(\mu_l \frac{dn_l}{dE} + \mu_h \frac{dn_h}{dE} \right) e E \quad (2)$$

Since the number of carriers is constant $\frac{dn_l}{dE} = -\frac{dn_h}{dE}$ and hence

$$\begin{aligned} \frac{dJ}{dE} &= (n_l \mu_l + n_h \mu_h) e + (\mu_h - \mu_l) e E \frac{dn_h}{dE} \\ \frac{dJ}{dE} &< 0 \quad \text{if } (n_l \mu_l + n_h \mu_h) < (\mu_l - \mu_h) E \frac{dn_h}{dE} \end{aligned} \quad (3)$$

Such a condition is favoured by high E , $\mu_l \gg \mu_h$ and by strong interband scattering processes to increase n_h when a strong field is applied.

It has been observed that in this regime the electric field distribution does not remain stable and gives rise to travelling field domains. Thus, so called high field domains periodically propagate across the crystal with the drift velocity of the carriers. The frequency ν of the oscillations associated with this motion is

$$\nu = v/d \quad (4)$$

where v is the velocity of the carriers and d is the thickness of the crystal slab. The typical values of $v \approx 10^7 \text{ cm s}^{-1}$ and $d = 10 \text{ micron}$ give $\nu = 10 \text{ GHz}$ (microwave X-band region).

(a) I-V Characteristics of Gunn Diode

Set the Gunn assembly micrometer at 5 mm. Disconnect the terminal from PIN modulator (BNC connector male (M) end) and keep the PIN bias knob full CCW. Connect the Gunn power supply to the Gunn diode through a BNC (M) to BNC (M) connector of Gunn bias. Set the voltage control knob to full counter clock wise (CCW) i.e. minimum Gunn bias and selector switch to current position. Now switch on the power supply, the display would indicate Gunn current. **If the current displayed in the digital panel meter (DPM) at any moment exceeds 850 mA switch-off the power supply immediately. The Gunn diode might have gone defective.** Note the current reading against the voltage. Change the Gunn bias in steps of 0.1 V by rotating the voltage control knob gradually clockwise. Read the Gunn bias voltage with digital voltmeter. Make use of a BNC T-connector. Plot the I vs V curve. The curve would look like as shown in Fig. 1.

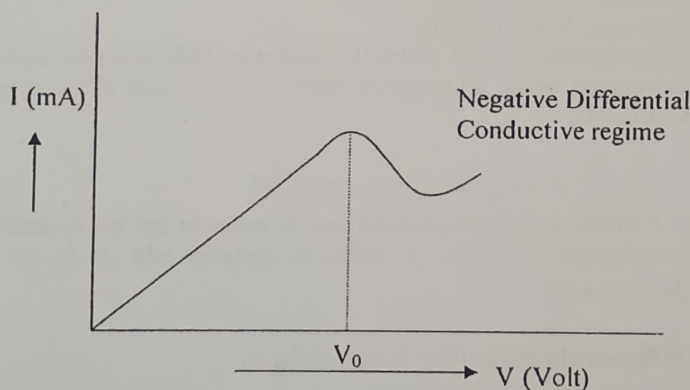


Fig. 1

Now change the Gunn assembly micrometer to another position (say $\sim 8\text{mm}$) and repeat the I-V data for this setting. Determine the threshold voltage V_0 . (The Gunn diode should not be biased at V_0 for longer periods it would burn the Gunn diode. Remember to minimize the voltage (full CCW position) if readings are not being taken. **KEEP PIN BIAS CONTROL KNOB FULL CCW (MIN).**

(b) Gunn as a Source of Microwaves

Before switching on the power supply ensure that the GUNN and PIN bias control knobs on it are turned full CCW (minimum). Connect the Gunn and the PIN diodes to the power supply using the corresponding connecting cables. Connect the slotted section wave

guide detector probe to indicator (a VSWR meter or a micro-ammeter or a CRO). The experimental setup is shown in Fig. 2.

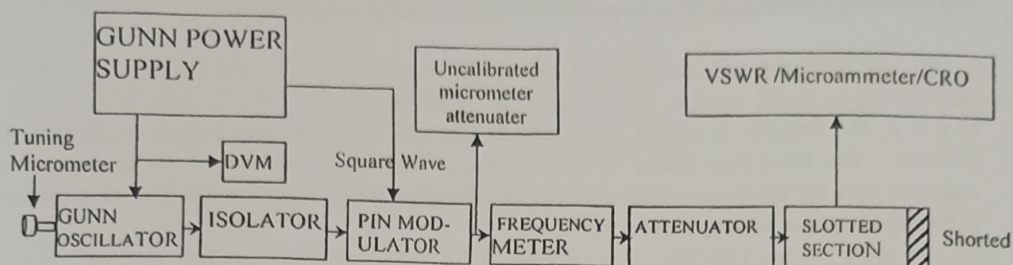


Fig.2 Experimental Setup for GUNN OSCILLATOR

Switch on the Gunn power supply, VSWR and CRO. Set the modulation selector knob on the Gunn power supply to MOD position. Gradually increase the Gunn bias to reach the negative differential conductivity regime (determined in section (a) above). Increase the PIN bias to an intermediate level (knob set midway between full CCW and full CW positions). If the settings are OK, you should observe square waves on CRO screen. The square waves are the amplitude modulated microwaves. The modulation is achieved by a PIN diode whose resistance depends on bias and its polarity. Therefore, a square wave voltage applied to PIN diode will produce a high resistance and a low resistance condition of PIN diode which in turn modulates the amplitude of the microwaves propagating through the wave guide section fitted with the PIN diode. Notice the Gunn assembly is separated from PIN assembly by an isolator. Adjust the position of the probe on the slotted section for the maximum reading in the indicator. Maximize the indicator reading by tuning the modulation frequency (frequency knob on Gunn power supply). Note down the Gunn bias and determine the frequency of the microwaves with the help of frequency meter. Change the micrometer reading of the Gunn oscillator and note the new frequency of the microwaves. Plot a graph between micrometer reading and frequency for a given Gunn bias. Repeat for another value of Gunn bias.

(c) Characteristics of PIN modulator

Adjust the Gunn bias and Gunn assembly micrometer to tune to a suitable frequency, set the attenuator ~3 dB (note it). Set the CRO input to DC. Adjust the time base and vertical gain to get a suitable square wave pattern with almost full scale vertical deflection. Read the PIN bias with the help of CRO. Make use of a BNC T-connector. Read the maximum (V_{max}) and minimum (V_{min}) voltages of the modulated microwave pattern on the CRO. The bottom (zero) level may be identified by grounding the input channel. Now determine the % modulation by the following relation.

$$\% m = \frac{V_{max} - V_{min}}{V_{max} + V_{min}} \times 100$$

Plot a graph between PIN bias and m .

(d) To Study the Response Law of the Detector

The average current in a microwave detector depends on the incident r.f. voltage. Let the current (I) be related to voltage (V) by the following response law

$$I = kV^n \quad (5)$$

If $n = 2$, the detector is called a square law detector and will detect the microwave power directly. We have here IN23 or IN23B crystal detectors and we wish to determine the response law for these microwave detectors. Same detector can show a different response in different ranges of input power levels.

Method (i): Connect the detector under test to VSWR. Set the VSWR gain near maximum. Move the detector probe along the slotted section to maximize the VSWR deflection. Keep the deflection within scale by increasing attenuation as required all the time. Adjust the fine gain of VSWR and attenuation to get the full-scale deflection of VSWR on dB scale. Note down the attenuator reading and the corresponding reading of VSWR on dB scale by increasing attenuation in small steps. Plot on a linear graph paper the VSWR reading vs. attenuator settings (attenuator reading number decreasing to right on x-axis and VSWR reading on y-axis number increasing upward). Since the attenuator reading represents the log of power (i.e. V^2) and the VSWR reading represents the log of detector current the curve should be linear for a square law region. Determine the slope of the curve. Repeat the experiment at several VSWR gains in decreasing order. You will see that the square law is obeyed for low power levels and the response may deviate at high power levels. **This method is done with the square modulated microwaves.**

Method (ii): This method is done with continuous wave (CW) microwaves i.e. **unmodulated waves**. For this the modulation is kept off i.e. change the modulation switch on the Gunn power supply to CW position and disconnect the PIN bias cable from the PIN diode. In this method, the current of the detector is measured by connecting the detector to a micro-ammeter. If the slotted section is shorted by a metal the electric field of the standing wave between two maxima in the slotted waveguide section varies as:

$$E = E_m \sin\left(\frac{2\pi}{\lambda_g} x\right) \Rightarrow E_m \sin\beta x \quad (6)$$

$$\text{where } \beta = \frac{2\pi}{\lambda_g}$$

where λ_g is twice the distance between two successive maxima or minima and x is the distance from the reference maxima or minima. Since the input voltage at the detector is proportional to the electric field at the probe it may be represented by

$$V = V_m \sin\left(\frac{2\pi}{\lambda_g} x\right) \Rightarrow V_m \sin\beta x \quad (7)$$

Therefore, first aim is to determine λ_g . This can be done easily by plotting current Vs. probe position to observe a number of maxima/minima. Now select a convenient

minimum and note the corresponding current. Move towards a maxima gradually on the graph. Note down the distance x moved by the probe and the corresponding change in detector current (I). Plot the graph on a log-log graph paper between I and $\sin\left(\frac{2\pi x}{\lambda_g}\right) \Rightarrow \sin \beta x$. The slope of the curve will give the exponent of the response law. For a square law detector the exponent should be 2. Repeat this for several levels of power (set attenuator at various settings between 0 dB and 10 dB).

(c) Calibrate the micrometer attenuator

Set the calibrated attenuator to 0 dB. Obtain the modulated square wave on CRO. Adjust the vertical gain to keep the pattern on scale (Full vertical deflection). Note the top level of square wave. Increase the attenuator value from 0 dB in known steps with the help of micrometer attenuator. Note the corresponding top level of square wave pattern. Now bring back the attenuator to 0 dB and produce the same corresponding change in the level of the top of the square wave. Note the corresponding micrometer reading of the uncalibrated attenuator and the corresponding attenuator reading of the calibrated attenuator. Plot micrometer reading Vs. attenuator reading. This plot gives us the calibration of the micrometer attenuator.

Questions:

1. Define decibel (dB). What do you mean by 3 dB attenuation? What is 80 dB noise level?
2. Estimate the thickness of GUNN diode used in the set up from its I-V characteristic curve assuming a threshold electric field of the order of 36 kV/cm is required to drive it towards negative differential conductivity regime.
3. Can the GUNN bias be used to amplitude modulate the microwaves? If yes how?
4. Why is it necessary to place an isolator or attenuator between a GUNN oscillator and PIN modulator?
5. Why is the slot in the slotted section of the waveguide located in the centre of the broad wall of the rectangular waveguide? What will happen if the slot is made in the narrow wall?

References:

1. Introduction to Solid State Physics – C. Kittel 5th Ed. Chap. 8, p. 245.
2. Introduction to Microwave Theory and Measurements, A.L. Lance, McGraw Hill Book Company.
3. E.M. Theory by E.M. Tralli.
4. Electromagnetic Waves and Radiating System – E.C. Jordan and K.G. Balmain.
5. Introduction to Microwaves: G. Wheeler
6. Introduction to Microwaves: H.A. Atwater.